

Project Design Phase

Introduction

The **Project Design Phase** is an important stage in the development of **EduGenie – Google Gemini Powered Learning Assistant**. In this phase, the overall structure, user interface, system components, and workflow of the application are planned. The design acts as a blueprint for developing the project.

1. System Architecture

EduGenie follows a simple architecture consisting of:

- **User Interface:** Allows students to enter their questions and view answers.
- **Application Layer:** Receives and processes user queries.
- **Google Gemini API:** Understands the question and generates a suitable response.
- **Database:** Can store user information, questions, and conversation history if required.
- **Response Module:** Displays the generated answer to the student.

2. User Interface Design

The interface is designed to be simple and easy to use. It may contain:

- Home page
- Login/Register page
- Chat or doubt-solving page
- Text input box
- Send button
- Answer display area
- Conversation history
- Help or information section

3. System Workflow

The basic workflow of the system is:

Student → Enters Question → EduGenie → Google Gemini → Processes Question → Generates Answer → Answer Displayed to Student

This workflow allows students to interact with the AI assistant in a conversational manner.

4. Module Design

The project can be divided into different modules:

- **User Module:** Handles student registration, login, and basic user information.
- **Doubt Solving Module:** Accepts questions from students and sends them to the AI system.
- **AI Processing Module:** Uses Google Gemini to understand the question and generate an appropriate response.
- **Response Module:** Displays the generated answer in a clear and readable format.
- **History Module:** Stores and displays previous questions and answers when required.

5. Database Design

If a database is included, it can store details such as:

- User ID
- Student name
- Email
- Question
- AI response
- Date and time

6. Expected Outcome

The Project Design Phase provides a clear blueprint for implementing EduGenie. A well-planned design helps developers understand how different components interact and makes the system **simple, organized, user-friendly, and efficient**.